CHM111 — Introductory Lecture

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Southern Union State Community College

CHM111

Warning:

This introductory lecture is only a summary of the syllabus; it is not a substitute for the syllabus.

The syllabus contains more details and is the official document that governs the course.

Please read it carefully.

Put your Technology Away.

- Phones are forbidden in class.
- No laptops, tablets, earbuds, headphones, etc.

If you need to use a phone for an emergency,
you may step outside the classroom to do so.

Reward for Adherence: "Active-Learning Points"

Outline

Part I:	Instructor Information
Part II:	Course Information
Part III:	Course Philosophy

Part I

Instructor Information

About the Instructor

Dr. Deskins

- Professor of Physics & Chemistry at SUSCC
- Husband and father

About the Instructor

Dr. Deskins

Education

- Ph.D. Physics, Auburn University
- M.S. Physics, Auburn University
- M.S. Chemical Engineering, University of Maryland
- B.S. Chemical Engineering, University of Maryland

About the Instructor

Dr. Deskins

Personal History

- Born and raised in Baltimore, Maryland
- First-generation college student
- Father: Navy veteran, industrial mechanic
- Mother: Homemaker, assistant in a library

About the Instructor

Dr. Deskins

Professional History

- Graduate Teaching Assistant
- Instructor: American International School in Beijing
- Instructor: SUSCC since Summer 2022

About the Instructor

Dr. Deskins

Axioms I Keep in Mind

- Carelessness Kills
- Hope for the Best, Prepare for the Worst
- The Enemy Gets a Vote
- Trust, but Verify
- No Bad Classes, Only Bad Teachers

About the Instructor

Dr. Deskins

Office Hours

- Office hours are used for retakes and questions.
- See my Work Schedule on Canvas for times.

Contacting Me

- Use **Canvas Inbox** (please never e-mail me)

Part II

Course Information

Outline of Part II: Course Information

Course Overview

Course Policies

Outline of Part II: Course Information

Course Overview

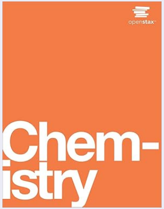
Course Policies

Course Description

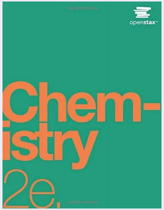
- College Chemistry I
- Lab required
- Prerequisites: MTH 112 and English competency

Course Textbook

Buy a used, physical copy of either of these. Either is acceptable. Do NOT buy ”Atoms First”.



Chemistry. OpenStax, 2016. ISBN-10: 1938168399.



Chemistry 2e. OpenStax, 2019. ISBN-10: 1593995784.

Buy a Used, Physical Copy of the Textbook

- Studies show reading comprehension is much worse when studying from a screen (Altamura et al., 2023).
- Buy a used, physical copy of the textbook for less than $10. This will save you time long term, as studying from a real book is more efficient.

Click here to open a search page on eBay.

Course Topics

- Matter, Measurement, & Problem Solving
- Atoms, Molecules, & Ions
- Stoichiometry of Chemical Reactions
- Aqueous Reactions & Solution Stoichiometry
- Thermochemistry
- Electronic Structure & Periodic Trends
- Chemical Bonding & Molecular Geometry
- Gases
- Thermodynamics & Intermolecular Forces
- Solutions & Colloids

What Makes This Course Different

Retakes

- **Unlimited retakes** on the 6 unit exams
- You keep your **highest score**
- No penalty for retakes
- Retakes are available until the end of the semester

Laboratory Work

- Labs are in person on designated days.
- See the Class Schedule for lab days.

Lecture Procedure

1. Students remove all technology and put the following on their desk:
   - pen/pencil
   - notebook
   - calculator
   - equation sheet
2. Instructor checks attendance.
3. Instructor distributes a sheet containing the lecture's objectives.
4. Instructor begins lecture.
5. Students raise their hand to ask questions as needed.
6. Instructor gives breaks periodically.
7. Instructor ends class and dismisses students.
8. Students with questions remain.

Symbols Commonly Used in Lecture

$\propto$	alpha (proportional to)	one quantity varies in proportion to another
Δ	delta (change in)	denotes a finite change in a quantity
$\therefore$	therefore	indicates effect from a cause
$\because$	since	indicates cause leading to effect
$\rightarrow$	implies	implies leads to / results in
$\sim$	scales with	indicates similarity or scaling
$\approx$	approximately equal	nearly equal in value

Emojis Commonly Used in Lecture

- 🔑 Key Takeaway or Main Idea
- 💡 Helpful Suggestion
- ☠️ Warning or “Don’t Make this Mistake”
- 🔬 Extra Detail or Deeper Explanation
- 📝 Important Note or “Write this Down”
- 📖 Reference to the Textbook
- 🪖 Problem-Solving Strategy
- 🎗️ Reminder or “Remember This”

Grades

Grading Breakdown

- Homework: 10%
- Unit Exams: 60%
- Final Exam: 30%

Letter Grade Scale

- A = 89.50–100.0%
- B = 79.50–89.49%
- C = 69.50–79.49%
- D = 59.50–69.49%
- F = 0.000–59.49%

Policies

- Final grade is the Canvas grade rounded to the nearest integer
- No curve; grades are honest.

Assessments

Exam Format

- One unit exam per module (6 exams in total)
- 15 free-response questions per unit exam (may be fewer on retakes)

Time Limit

- Questions are written to take 4 minutes on average ($\therefore$ 1 hr for 15 questions)
  - I give as much time as I can, but there is a limit (e.g. end of class or day)

Work Must Be Shown

- All major steps must be shown to receive credit.
- Correct answers with no or low-quality work will receive no credit.

What Does an Exam Problem Look Like?

Example Problem with Solution

Caleb drives his 1160-kg car around a 210-m radius unbanked curve at 36.8 m/s (about 82 mph). Caleb's mass is 64 kg. Calculate the centripetal force exerted on the car as Caleb drives around the curve.

ANSWER: 7.89 kN

$$a_c = \frac{v^2}{r}$$

$$F_c = ma_c$$

Write the equation(s) as seen on equation sheet

$$F_c = \frac{mv^2}{r}$$

Use algebra to solve for variable of interest

Plug in values (with units)

$$F_c = \frac{(1160\,\text{kg} + 64\,\text{kg})(36.8\,\frac{\text{m}}{\text{s}})^2}{210\,\text{m}} = 7893\,\frac{\text{kg}\,\text{m}}{\text{s}^2} = 7893\,\text{N} = \underline{7.89\,\text{kN}}$$

Report final answer

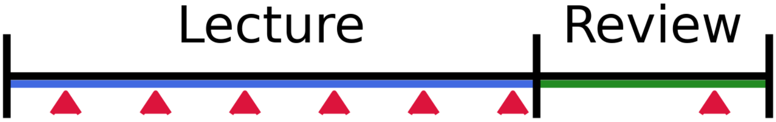
☆ Correct answers with no or low-quality work receive no credit

SHOW YOUR WORK!

Exam Procedure

1. Students remove all technology and put the following on their desk:
   - pen/pencil
   - calculator
2. Students spread out.
3. Instructor distributes exams and equation sheets.
4. Students with questions raise their hand.
5. Students going to the restroom must show their phone is left in classroom.
6. Student submits their completed exam and sits down to await its return.
7. Instructor promptly grades the exam and returns it with an answer key.
8. Student leaves.

Course Schedule



- Lectures End Before Semester Ends $\because$ Extra Time $\Rightarrow$ Procrastination $\therefore$ Waste
- Between Lecture End and Semester End: Retakes and Guided Review
- Retakes for an exam are available after that exam is given.
- I *try* to allow retakes in the days following the final exam, before grades are due.

Steps for Seeking Help

- You should follow the following steps for seeking help.
- If you do not reach a level of clear understanding, proceed to the next step.

Procedure for Overcoming Failures in Understanding

1. **Read the textbook.**
2. Ask an AI model.
3. Ask your peers.
4. Ask me.

Every student experiences difficulty at some point.

Here are the steps to follow if you need help with something. Go in order and don't skip #1.

Expectations of Help During Office Hours

- Office hours are used for retakes and for seeking help when other methods fail.

Course Grade $\leq$ 35%

- "Training wheels" are likely helpful for you to improve.
- You can expect me to give you a problem and provide a solution to you and walk you through it, step by step, checking for understanding to get you to a state of independence and self-reliance.

Course Grade $>$ 35%

- "Training wheels" are likely harmful for you to improve.
- Try your best to rely on yourself.
- Bring me your own solution attempt and ask me clear questions.
- I may hand you the textbook if I sense a need to read the chapter.

Outline of Part II: Course Information

Course Overview

Course Policies

Attendance & Deadlines

Expectations

- Attend all class meetings. Arrive on time and stay until class is dismissed.

Natural Consequence of Absence

- Absence $\Rightarrow$ lower grade; I impose no punishments for absences.

No Late Work

- Submit work on time; no late work is accepted.

Academic Integrity

Intolerance

- Cheat on an exam $\Rightarrow$ automatic F in the course

Technology in the Classroom

- No technology is allowed in the classroom.
- This includes but is not limited to phones, laptops, earbuds, and tablets.

- Slides are available online, so you may not take photos during class.
- Sit near the front if visibility is an issue for you.

Why? Because this is a difficult course even without distractions.
Do not make it more difficult by distracting yourself and others.

Artificial Intelligence (AI)

Beneficial Use of AI

- Seeking explanation of your mistakes

Detrimental Use of AI

- Using AI to solve problems and memorizing solutions
- Overreliance on AI; Excessive confidence in AI

Excused Absences

Excused-Absence Policy

- Absences are excused for valid reasons: (1) Illness; (2) College-sponsored activities; (3) Death in immediate family; (4) Military duty; (5) Jury duty / court appearance.
- You must notify me as soon as possible and provide documentation for the excused absence.

Little Need for this Policy in this Course

That being said, since I offer unlimited retakes on exams, you can simply take a missed exam when you are able to and do not need to worry about excused absences.

In Case of Illness

Do not come to class if you are feeling ill.

Stay home in order to

- rest and recover faster
- prevent others from falling ill

Withdrawal

Withdrawal Policy

- While I never recommend it, you can withdraw from the course and receive a "W" grade before a certain date (see syllabus). GPA is not affected by a "W".
- After the withdrawal deadline, you will receive a letter grade based on your performance in the course.
- If you are struggling in the course, please reach out to me ASAP and we can discuss strategies for improvement.
  My job is to teach you chemistry; if you are not learning, then I am failing.

Part III

Course Philosophy

Why spend valuable time discussing course philosophy?

- sets expectations early: if you understand how to approach the course well, studying is more efficient.
- improves resilience and student success: understanding the course philosophy makes it more likely that you will work through difficulty instead of quitting.

Outline of Part III: Course Philosophy

Value of Education

Grade Inflation

Learning Strategy

Success through Failure

Value of Education

Education is meant to be pursued primarily for the cultivation of wisdom and the appreciation of truth, goodness, and beauty; career success may follow, but it should remain secondary rather than the ultimate aim.

> *"Πάντες ἄνθρωποι τοῦ εἰδέναι ὀρέγονται φύσει."*
>
> — Aristotle
>
> Metaphysics, Book I, Part I

φῠ́σῐς [pʰý.sis] *noun* – Ancient Greek

nature, quality, property

Note: φῠ́σει is φῠ́σῐς in its dative-instrumental form, hence the appearance of "by" in the translation.

All men desire to know by nature.

φύσις ⇒ φυσική ⇒ physica ⇒ physics

Greek ⇒ Latin ⇒ English

(I know this is chemistry, not physics, but bear with me.)

Value of an Activity

Do YOU feel sometimes feel guilty when you spend time...

- watching TV?
- playing video games?
- scrolling through social media?
- engaging in a hobby? (e.g. rock-collecting, gecko-breeding, bonsai, skydiving)

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We can feel guilty when engaging in apparently **useless** recreational activities.

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We can feel guilty when engaging in apparently **useless** recreational activities.

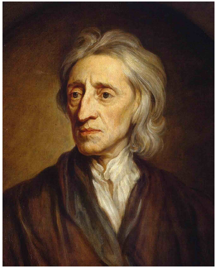
Why?

Role of Leisure

Aristotle's Moral Philosophy (4th Century BC)

- Leisure $\Rightarrow$ human flourishing ($\therefore$ is the best use of time).
- Work (productive labor) is only needed to secure time for leisure.
- Play is to be used only to relieve temporary stress. The pursuit of endless pleasure with the goal of happiness is a childish fantasy.

Role of Leisure



John Locke's Moral Philosophy (17th Century)

- Work (productive labor) is the primary act of moral value
  $\therefore$ time is best spent working and being productive.
- Non-working time is justified as an instrument: rest and play are permitted insofar as they restore us for more work.
- Leisurely acts are actually acts of work, striving for X instead of appreciating X (e.g. collecting rocks exercises practical reason).

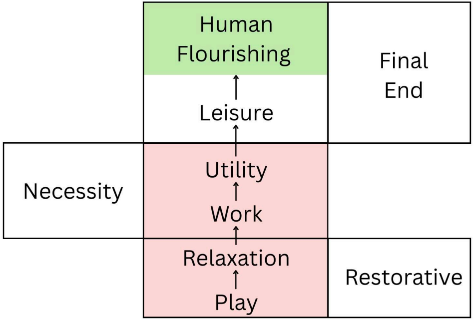
Aristotle v. Locke

- Aristotle: leisure is the highest human activity because it orders us toward truth, beauty, and contemplation.
- Locke: productive labor is primary; leisure is acceptable mainly as recovery for more work.

The Foundation of the Disagreement

- Aristotle: things are good when *they are fitting for human flourishing.*
- Locke: things are good when *they are useful.*

Modern culture largely treats usefulness as the highest standard of value.



Hierarchy of Acts

- Leisure: most valuable (end in itself).
- Work: only valuable as a means to leisure
  $\therefore$ work is inferior to leisure.
- Work creates the need for relaxation, which is achieved through play and restores us so that we can work again.

Why do I get Bored on a Long Vacation?

- In an environment with work, restorative play is necessary.
- In an environment without work, play is unnecessary and is therefore boring.

"If you're not having fun, you're not learning. There's a pleasure in finding things out."

— Richard Feynman
Nobel Laureate (Physics), 1965

What is School?

σχολή [skʰo.lɛ̌ː] *noun* – Ancient Greek

leisure (meaningful non-work duties), free time

(also a place in which leisure time is spent, especially lecture, disputation, discussion, philosophy)

- Adults were expected to spend most of their free time on thoughtful, enriching leisurely acts, appreciating truth, beauty, and goodness.
- School meant to be leisure, not work (and certainly not play).
- By being curious, you find pleasure in learning and discovering new things.

What is School?

- Purpose of a video game: to amuse (useful insofar as to relieve stress).
- Purpose of a piano: to empower understanding $\Rightarrow$ appreciation for beauty.
- Purpose of chemistry: to empower understanding $\Rightarrow$ appreciation for truth.

- Note: $\Rightarrow$ means “implies” or “results in” and is used to indicate a logical consequence of the preceding statements.

Role of Education in your Life

1. Man is naturally drawn to knowing and understanding.
2. Education is a disciplined form of that desire.
3. The highest goods of education are not external rewards, but the formation of the intellect and the appreciation of truth, goodness, and beauty.
4. Grades, jobs, and money are real goods, but are instrumental goods, not the final purpose of learning.
5. $\therefore$ you should view education as a search for wisdom and the appreciation of truth, goodness, and beauty, not merely as a requirement to begin your career.

Outline of Part III: Course Philosophy

Value of Education

Grade Inflation

Learning Strategy

Success through Failure

Problems Affecting College Students

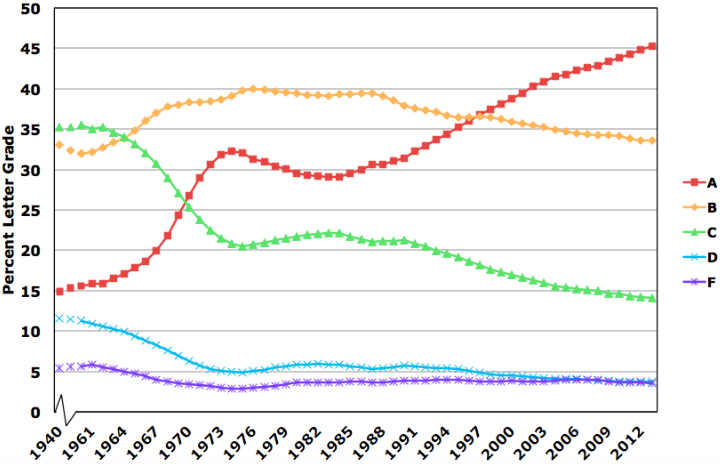
- I find that many of my incoming students are unprepared, often lacking skills in problem solving, math, and reading comprehension.
- Studies confirm a decline in student quality.

Findings from Scientific Studies

Many modern-day college students...

- lack emotional maturity (Fry, 2023)
- exhibit issues with entitlement and dependency (Knepp & Knepp, 2022)
- struggle with technology addiction ($\Rightarrow$ difficulty focusing) (George et al., 2024)
- do not devote the time to studying that is required (Flaherty, 2023)
- lack reading-comprehension and reasoning skills (Pudewa & Walker, 2024)
- lack resilience (Suhaimi et al., 2024)

Rising Grades, Declining Standards



- Average college grades have risen over time.
- Average student is less skilled than previous generations.
- ∴ Grades have become worse at measuring mastery.

Declining Standards to Conceal Declining Quality

Standardized tests and accreditation exams have also been made easier over time (Ryan, 2004) (Cobourne, 2023) (Yadav et al., 2023).

This further obscures the fact that modern-day students are less competent than students of previous generations.

Requirements of an Excellent Education

My duty is to give you an excellent chemistry education, which requires you to demonstrate true mastery of the concepts.

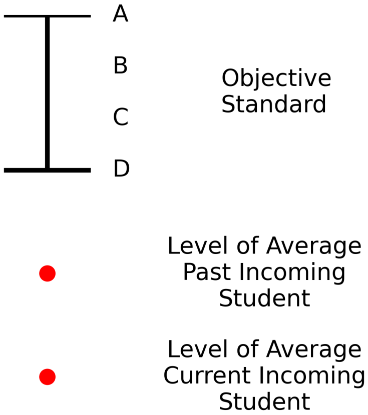
My Expectation: Concept Mastery

Since grades are inflated in many classes, many students now expect an "A" without actual mastery of course material.

Your Expectation: Good Grade

You and I must work together so that both are achieved.

Concept Mastery $\Rightarrow$ "A"

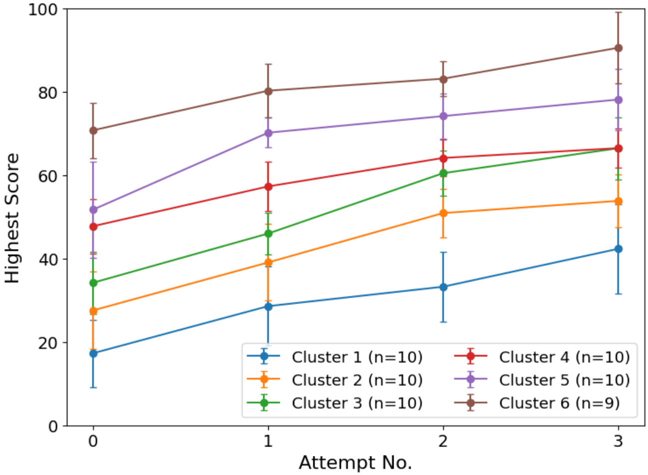


Current incoming students are less prepared for my class than their predecessors.

They still need to meet the same standard.

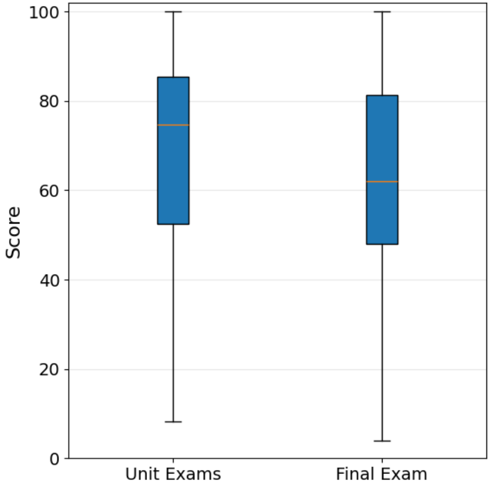
Unlimited Exam Retakes $\Rightarrow$ Better Learning

Results of Retakes



- Original Attempt is Attempt 0.
- Retakes increase understanding, which increases scores.
- Uniform increase across student clusters ($\sim$25% after 3 retakes).
- Studying for each retake naturally improves long-term retention, which means that studying for the final exam is easier.

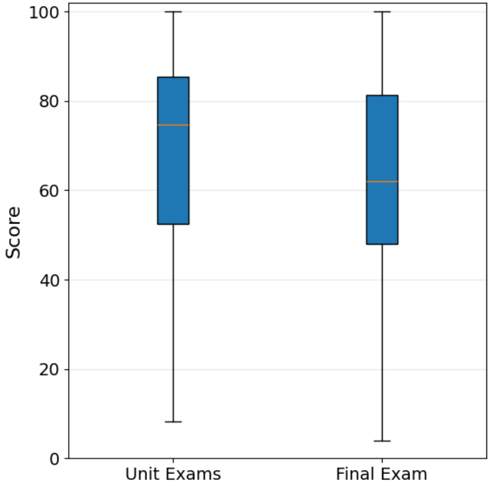
Results of Retakes



	Unit	Final
Mean	68.3%	60.2%
σ	23.4%	25.4%
Min	8.3%	4.0%
25%	52.6%	48.0%
50%	74.7%	62.0%
75%	85.5%	81.3%
Max	100.0%	100.0%

Summary statistics showing student performance. The set of students' averaged best scores on unit exams ("Unit") is shown alongside the set of final exams ("Final"). (N = 62 students)

Results of Retakes



- Score on final exam is predicted to be 8.1% lower than unit-exam average.
- Retakes are a form of frequent assessment (proven to increase long-term retention).
- Good long-term retention is clearly shown on the final exam.

Working Together to Master Concepts and earn an "A"

My Contributions to this Goal

- High-Quality Lectures
- Detailed Learning Objectives
- Homework with Solutions
- Labs Connected to Content
- Unlimited Exam Retakes
- Practice Exams with Answers
- Sample Exams with Solutions
- Past Student Exams with Solutions

Your Contributions to this Goal

- Studying
- Attention during Lecture
- Asking Questions
- Preparing for Exams
- Learning from Past Mistakes
- Self-Reliance in Studying
- Collaboration with Classmates

Outline of Part III: Course Philosophy

Value of Education

Grade Inflation

Learning Strategy

Success through Failure

How to Be Successful in this Course

1. Read the chapter in the textbook (make sure you have a physical copy).
2. Carefully follow the example problems.
3. Pay attention during lecture. (Ideally read the chapter before lecture).
4. Do the homework. Struggling? Go back to the chapter. Still struggling? Look up the solution on YouTube.
5. Review my sample exam. Try to solve the problems without the solutions.
6. Review past students' sample exams. These show good student-level work.
7. Do the practice exam. This has no solutions, only answers.
8. Do the exam. Review steps 5-7 and retake until you get 100%.

With high exam grades secured, only light review is needed for the final exam.

How to Be Successful in this Course

- Read the chapter in the textbook before lecture so that you can follow along more easily.
- Just because you miss the opportunity to read before lecture does not mean that you can skip reading; do the reading after lecture.
- Do not expect a good understanding from only attending the lecture.

The following is an alternative, less effective plan:

How to Be Successful in this Course

1. Pay attention during lecture. (Ideally read the chapter before lecture).
2. Read the chapter in the textbook (make sure you have a physical copy).
3. Carefully follow the example problems.
4. Do the homework. Struggling? Go back to the chapter. Still struggling? Look up the solution on YouTube.
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With high exam grades secured, only light review is needed for the final exam.

Beware of Blame. Beware of Overreliance.

- The **externalization of responsibility** for **controllable outcomes** leads to a **diminished sense of agency**, which subsequently reduces **motivation** and causes **dependency**.
- i.e. if you blame others for what is in your control, you deny your own power to improve the situation, lose motivation to try, and often must depend on others.
- **Blame** $\rightarrow$ **Learned Helplessness** $\rightarrow$ **Dependence on Others**

"To Teach" is to Guide

tǣcan [ˈtæː.t͡ʃɑn] *verb* – Old English

to show (someone) the way, direct, guide

- Your duty as a student: to learn and become educated.
- My duty as a teacher: to guide you in this learning process.

- A sherpa enables climbers to ascend by guiding their path.
- A teacher enables students to learn by guiding their inquiry.

- Learning is not passively accepting spoon-fed information.
- Learning is actively following the path shown by the teacher.

Outline of Part III: Course Philosophy

Value of Education

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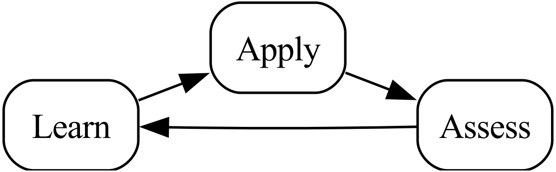
Failures of the Traditional Model of Schooling

Traditional schooling looks like this:

But real learning looks like this:



Learning things in a natural setting is a cycle.

Rarely are we given just one chance to demonstrate proficiency.

So then why are students just given one chance at an exam?

My retake system reflects how we actually learn.

Example of Natural Learning: Riding a Bicycle

- See how to ride.
- Try in a controlled setting to build confidence.
- Ride on your own.
- Fall.
- Get up, adjust, and try again.

- This is how mastery develops.
- Falling is a natural part of the learning process.

Zeus the Guide, who made man turn
Thought-ward, Zeus, who did ordain
Man by Suffering shall Learn.
So the heart of him, again
Aching with remembered pain,
Bleeds and sleepeth not, until
Wisdom comes against his will.
'Tis the gift of One by strife
Lifted to the throne of life.

— Αἰσχύλος (Aeschylus)
Ἀγαμέμνων (Agamemnon)
(458 BC)

Zeus made humans think and reflect.
He made it so that people must
learn by suffering.
So a person's heart,
aching with memories of pain,
doesn't rest or sleep
until wisdom comes,
even if they don't want it.
This is the gift of Zeus,
who rose to power through struggle.

Translation on the left by
Prof. Gilbert Murray
University of Oxford
(1920)

Failure as the First Attempt In Learning (F.A.I.L.)

- Everybody fails to grasp things while learning chemistry.
- Failures reveal what is not yet understood.
- Failures are useful when you learn from them.

What to do When you Fail

- Make an improvement plan and execute. Hope by itself is not a strategy.
- That being said, you do need hope. You have the power to improve. Get up and try again.

Why Failure Matters

- Learning is moving from ignorance to understanding.
- Facing the unknown begins the learning process.
- Struggling to comprehend the unknown forces thinking.
- Understanding is reached after struggling and questioning.

Beware the Illusion of Mastery

Beware: Reading over others' work can create a false sense of understanding. Do not assume you understand until you demonstrate mastery on an exam.

What Failure should Produce

- Better study habits
- More careful problem solving
- Stronger content knowledge
- Development of greater confidence after success

Humility → Wisdom

- Arrogance prevents improvement and invites failure.
- Arrogance makes one a fool in appearance and substance:
  1. Nobody believes the braggart; he only looks like a fool.
  2. One must admit ignorance to learn and improve. Since the arrogant think they know everything, they remain ignorant and foolish.

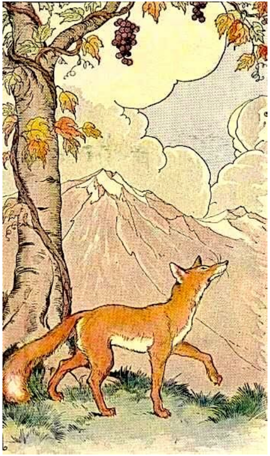
To become wise, you must truly believe yourself to be ignorant.

To remain ignorant, you must truly believe yourself to be wise.

Reflect on your Mistakes and Improve

- Some students habitually make excuses to cope with failure and lower stress
- But excuses become less believable as more exams are taken
- $\therefore$ retakes remove excuses and force accountability
- $\therefore$ retakes cause stress in students unaccustomed to honest self-reflection
- Stress $\Rightarrow$ fight-or-flight response
- Students with a growth mindset do retakes and improve
- Students with a fixed mindset stop attending and fail or drop out

The Fox and the Grapes



This Fox has a longing for grapes:
He jumps, but the bunch still escapes.
So he goes away sour;
And, 'tis said, to this hour
Declares that he's no taste for grapes.